KOR STRUCTURAL · ENGINEERING TOOLS

A structural model built from the drawings

31168 YMCA Langara — Tower B and the rest of the building · prepared 6 August 2026

Andrea's ETABS models were lost with a failed drive. Rather than re-enter the building's geometry by hand, this reads the concrete-outline DXF plans drafting already exports and produces an ETABS model covering every storey. It does none of the engineering — it removes the typing.

58 storeys populated	**757** wall panels, true thicknesses	**2,393** columns, sized & oriented	**102** floor plates, median 9,800 ft^2

What to open

FILE	WHAT IT IS
`31168-FROM-DRAWINGS.e2k`	The model. **File → Import → ETABS .e2k Text File.**
`31168-FROM-DRAWINGS-report.txt`	Every sheet, the storeys it landed on, what it produced, and every location that needed judgement.
`KOR-Model-From-Drawings-DOSSIER.pdf`	This document.

All of it sits in `02 Engineering\02 Lateral Design\01 ETABS Models\`. Everything generated is named `KW*`, `KC*`, `KP*` with sections prefixed `KOR-`, so you can select, filter or delete the whole lot in one action.

What has been verified, and what has not

Verified: every member sits exactly where the drawing draws it

Each plan is rendered with the extracted members drawn over it (`takeoff dxf-render <plan> <png> --overlay`), so what the model carries can be compared against the linework directly rather than inferred. On B-LEVEL 30 the core's two shafts, its bar wall and all twenty-four perimeter columns land on their outlines; the images are in `_drawing-renders\`. Wall thicknesses and column sizes are measured from the outlines, not assumed.

Not verified: where the geometry sits relative to your grid

The building has a second route into ETABS — the Revit analytical model through CSiXRevit — but it covers only a fragment of Building C, a different part of the site from the tower sheets, so it cannot confirm placement for most of the model. All generated geometry does fall inside the model's grid envelope, and the drawings and the CSiXRevit export both come from the same Revit project, so the coordinate frames should agree — but "should" is not "checked".

Please confirm this first. Open a storey in plan and check that the core and columns land on their grid intersections. If the geometry is offset, it will be offset by the same amount everywhere: select all and move it once, or tell us the offset and the model can be regenerated with `--offset x,y`. One storey on C-LEVEL 3 also looks like it may have taken geometry from the wrong sheet — worth a look before trusting the lower levels.

What it reads, and how

LAYER	BECOMES	HOW
`JBP_V-WALL`	Wall panels	A core is drawn as one ribbon tracing both faces of a group of walls. Each face is paired with the face opposite it; the pair gives the centreline and the true thickness.
`JBP_V_COL`	Columns	Footprint gives size and rotation, so a 16×24 sits the way it is drawn.
`JBP_C_SLABEDG*`	Floor plates	Outlines stitched into rings; rings inside a ring are openings.

Sheet titles carry the placement. `LEVEL 29 PLAN (L29-35)` fills seven storeys; `A-LEVEL 28` goes to Tower A alone; `LEVEL P2` to the parkade. The geometry is merged into an `.e2k` that ETABS itself exported, so storeys, grids, materials and design settings remain ETABS's own — the file differs from a known-good one only by the lines added.

Limitations — read before you rely on it

Slab plates are the weak half

Drafting interrupts slab edges wherever other linework crosses them, so many outlines do not close. Those are dropped and reported rather than guessed at, and rings under 50 ft^2 are discarded as noise — the 88 plates that survive have a median area of 9,700 ft^2, which is a real floor plate rather than a sliver. Coverage is therefore partial: the plates that are there are believable, but not every storey has one. Treat them as a starting point, or ignore them and place your own diaphragms.

Widening the closure tolerance was tried and rejected on the evidence: at 18″ the model gained 302 more plates, but the count of plates in a plausible size range did not change at all — every addition was a fragment. Recovering the missing plates needs a better method, not a looser one (see below), and that is why only one model is shipped.

Nothing is assigned that is yours to decide

No diaphragms, no loads, no masses, no spectra, no stiffness modifiers, no piers or spandrels, no meshing. One concrete material is applied to every generated section — reassign by element.

Openings are detected but not cut

Shafts and stair openings are found as inner rings and reported, but not subtracted from the plates, so slab areas read high where a core passes through.

> **Check the thick walls.** Anything modelled thicker than 24″ is flagged in the report. Wall outlines that meet at junctions can pair across the junction and read thicker than the wall really is. The flags tell you exactly where to look.

It believes the drawing

Thicknesses and sizes come from what was drawn, not from a schedule. A wall drawn at the wrong width models at the wrong width. The plans used here are dated 28 June 2026 — later revisions are not in this model.

What would make it materially better — your input decides

IMPROVEMENT	WHAT IT NEEDS FROM YOU
Cut the openings. Subtract shafts and stairs from the plates so slab areas and diaphragms are right.	Confirm that inner rings on the slab-edge layers are always openings, never a step or a depression.
Close slab edges properly by extending an interrupted edge along its own direction to the line that cut it, instead of bridging by distance. Measurement says this is the only way to recover the missing plates.	Nothing — but tell us which storeys you most need plates on, so it is proven where it matters.
Wall grades by height. Read the wall schedule so 65 MPa low and 45 MPa high come through instead of one material.	Point us at the schedule sheet you would read for this, and how you would expect the grades keyed to walls.
Beams. Nothing on the beam layers is read today.	Whether beams are worth having, or you would rather add them yourself.
A button in the KOR app instead of a command line.	Whether you would run this yourself per project, or ask for it.

The single most useful thing you can send back: open the model, and tell us the first three things that are wrong or annoying. Each one is a rule the tool does not yet know.

Where this sits in the KOR toolset

This is a new front end on the engineering engine the app already carries, not a separate program.

```
Revit  ──→  drafting's DXF plan exports  ──→  [ DXF → ETABS ]  ──→  .e2k  ──→  ETABS
PDF drawings ──────────────────────────────→ [ PdfToSafe    ]  ──→  .f2k / .e2k / live OAPI
PDF drawings ──────────────────────────────→ [ Takeoff      ]  ──→  quantities, xlsx
```

All three live in `Kor.Operations.EngineeringTools`, share the same geometry code, and are covered by the same test suite (384 tests). PdfToSafe reads drawings when only a PDF exists; this reads the DXFs when drafting has exported them, which is exact and needs no vision model. The natural next step is a button in the app's Engineering Tools panel so it runs without a command line.

> **It only stays reliable while the layer names hold.** `JBP_V-WALL`, `JBP_V_COL` and `JBP_C_SLABEDG` are now a machine contract, not just a drafting habit — which is a concrete argument for the Revit template and standards work already under way.

Built 6 August 2026 · `Kor.Operations.EngineeringTools.Core/Dxf` · CLI `takeoff dxf-to-etabs`, `dxf-inspect`, `e2k-compare` · documented at `docs/DxfToEtabs.md`